DEV-104 | Samarth Life Management | Hackathon PRD v1.0

Samarth Sentinel: an early-warning system for seniors living at home

Tell a Care Coordinator **who needs attention today, why, and what to do next**. Flag risk and route it to a human, never diagnose.

Hindi + English | Voice-first for seniors | Explainable | Low cost

1 Problem & opportunity

Decline in older adults is gradual and noticed late. Each signal is small; together they predict trouble.

149M

Indians aged 60+ in 2022 (10.5% of population), projected to reach 347M (20.8%) by 2050.
UNFPA India Ageing Report 2023

~1 in 5

older persons live alone or only with a spouse and manage needs on their own.
UNFPA, 2023

24% · 23%

face limits on daily activities · have two or more chronic ailments.
UNFPA, 2023

40,000+

seniors across 350+ cities supported by Samarth, but coordinators prioritise from memory and notes.
Problem statement

Insight. The bottleneck is *attention allocation*, not data scarcity. The product is a ranked, explained, low-noise daily worklist, so a coordinator can act on it tomorrow.

2 Goals & non-goals

Goals

- Catch decline days or weeks earlier than today’s process
- Rank seniors daily with plain-language reasons and one concrete next action
- Keep alert volume low so coordinators trust it
- Let low-tech seniors check in by call or voice, in Hindi or English
- Run affordably at 10,000+ seniors

Non-goals

- No diagnosis, no condition names to seniors or families
- No replacement for emergency services or clinicians
- No senior-facing app
- No real patient data in the prototype

3 Users & needs

🧑‍💼 Care Coordinator

Ranked list, reasons, suggested action, one-tap feedback. **Primary user.**

👴 Senior

Basic phone. Missed-call check-in, IVR keypress or a voice note in Hindi/English.

👨‍👩‍👧 Family member

Short weekly wellbeing note, immediate alert only if serious, with consent.

📊 Operations manager

Alert volume, response time and outcomes by city or team. Aggregates only.

4 Functional requirements

P0 must ship in demo · P1 should · P2 stretch

ID	Requirement	Priority	Acceptance check
FR-1	Ingest vitals (BP, sugar, SpO2, weight), medication logs, call notes (text/voice); activity and family input optional	P0	3+ data types per senior, tolerates missing entries
FR-2	Extract health cues from Hindi/English/ Hinglish notes: pain, fall, confusion, low appetite, loneliness, dizziness, poor sleep; handle negation and dates	P0	Per-cue precision/ recall on held-out labelled notes
FR-3	Daily risk score per senior: rules baseline + per-senior baseline anomaly + ≥1 ML model	P0	Ablation shows each layer adds value
FR-4	Coordinator dashboard sorted by urgency with top reasons and recommended action	P0	Every alert shows ≤3 plain-language reasons + 1 action
FR-5	Alert controls: daily cap, dedupe, cool-down, escalation on persistence	P0	Alerts per coordinator-day within target
FR-6	Role-based access and consent-gated family sharing; PII masking before any external call	P0	Tests prove no cross-role leakage or unmasked PII
FR-7	Ops view: volumes, response times, outcomes	P1	Aggregate view per city/team
FR-8	Weekly family summary in warm, non-diagnostic language	P1	Banned-phrase guardrail passes 100%
FR-9	Coordinator feedback (useful / not useful) tunes thresholds	P1	Precision improves on replay

FR-10	Voice check-in bot (3–4 Hindi questions, DTMF fallback), inactivity/fall from sensor data, missed-dose patterns	P2	Simulated flow end to end

5 Intelligence design

Signals → **Cleaning & features** → **NLP cues** → **Rules + baseline + ML** → **Alert engine & explainer**

Layer A · Rules

Transparent safety nets: repeated missed doses, low SpO2, rapid weight change, fall keyword. Auditable and always on.

Layer B · Own-baseline

Rolling robust deviation (median/MAD) per senior over 14–28 days, plus slope and *silence* features. Own history beats population averages.

Layer C · ML

LightGBM on windowed features with SHAP; Isolation Forest/changepoint as unsupervised complement. Small models fit ~200×90 data.

NLP

Hinglish lexicon + small encoder classifier (HingBERT-class), script normalisation, negation and time handling. LLM only for hard cases, on masked text.

Example alert (plain language, no diagnosis)

🔴 **High · Sunita D.**
Why: BP ~18 mmHg above her usual for 4 days · missed 3 of last 5 evening doses · last two calls mention dizziness and low appetite.
Do next: call today, confirm medicines, suggest doctor visit if it continues.

🟠 **Medium · Ramesh K.**
Why: calls unanswered 3 days (usually answers) · sleep complaints rising.
Do next: ask family to check in; retry call this evening.

Safety language rule. Output may say “needs attention”, “has been unwell-sounding”, “suggest a call”. It must never name a condition or say “you have…”. A banned-phrase test runs on every message.

6 Experience requirements

Coordinator dashboard

- Top-N list by urgency, capped daily
- Reason chips + trend sparkline with baseline band
- Actions: call, log outcome, snooze, escalate
- 👍/👎 feedback per alert

Senior channels

- Missed-call check-in
- IVR: 1 = fine, 2 = not well, 3 = need help
- Voice note → Hindi/English speech-to-text
- Slow, short, repeated prompts

Family

- Weekly warm summary
- Instant alert only if serious
- Consent toggles per type

Ops

- Alerts/day, time-to-first-action
- Outcome mix, false-alert rate
- Aggregates, no raw health data

7 Architecture & stack

Layer	Choice	Why
Backend	Python, FastAPI	One language for API, ML, NLP
Auth & DB	Supabase (Postgres, Auth, RLS, Storage)	Access rules enforced in the database; private audio buckets
ML	pandas, scikit-learn, LightGBM, SHAP	Right-sized, explainable
Speech	Open Hindi Whisper-family models (self-hosted)	Privacy and low cost

Dashboard	Streamlit or Next.js	Choose by team skill
Deploy	Docker Compose on one small VM	One-command judge setup

Supabase notes. Free tier includes 500 MB database and 2 projects, and free projects pause after a week of inactivity, so keep the demo project warm. Pro is $25/month and stays running. ML jobs use the service-role key server-side only; enable RLS on every table. Verify current limits and the Mumbai region before launch.

8 Privacy, safety & compliance

Controls

- Roles: coordinator, family, ops, admin
- Consent flags per family link and purpose
- PII masking layer + test before any external API
- Audit log of every view
- Short audio retention

India’s DPDP Rules

Notified in November 2025 with a phased rollout: Board provisions first, consent-manager rules from 14 Nov 2026, and the main duties (notice and consent, breach reporting, safeguards) from 14 May 2027. Seniors under legal guardianship need a guardian-consent path. Law-firm summaries; not legal advice, get counsel before real data.

9 Success metrics & evaluation

Targets below are proposed starting points to tune on the synthetic cohort, not published benchmarks.

≥ 80%	≥ 60%	≥ 5 days	≤ 10
episode recall on planted declines (alerted before event)	precision@10 per coordinator-day	median lead time before event day	alerts per coordinator per day (cap)

- **Cohort:** ≥200 seniors × 90 days, 10–15% planted declines with known onset, plus hard negatives. Split by senior, validate on a fresh seed.
- **Ablations:** rules only → +baseline → +ML → +NLP cues.
- **Stress:** higher missingness, noisier notes, ASR errors.
- **Honesty:** synthetic results show the pipeline works, not clinical efficacy.

10 Roadmap

Phase 1 · Core (hackathon)
Synthetic generator, ingestion, rules + baseline, evaluation harness, coordinator dashboard.

Phase 2 · Intelligence
NLP cue extractor, LightGBM + SHAP, alert caps, privacy tests, ops view.

Phase 3 · Polish & stretch
Family summary, feedback loop, simulated Hindi voice check-in, write-up and pitch.

Post-hackathon pilot
Real-data shadow mode with one coordinator team, consent flows, legal review, threshold calibration.

11 Risks & mitigations

Risk	Impact	Mitigation
Circular evaluation on own synthetic data	Inflated metrics	Different seed and scenario parameters for validation; hard negatives; state limits
Alert fatigue	Coordinators ignore system	Daily cap, cool-downs, feedback loop, precision@K focus
Noisy phone ASR on Hinglish	Missed or wrong cues	Confidence flags, human review, DTMF fallback
Accidental diagnostic wording	Safety and trust	Template-only messages, banned-phrase tests
PII leaks to external services	Privacy breach	Masking layer, self-hosted models, tests

12 Research sources & models to cite

- UNFPA & IIPS, *India Ageing Report 2023* (population and living-arrangement figures).
- MeitY, Digital Personal Data Protection Rules 2025 (law-firm summaries consulted).
- AI4Bharat IndicWhisper / Vistaar (Hindi ASR); Bodhan AI Indic-Transcribe (announced September 2026, verify licence); Whisper-Hindi2Hinglish (Oriserve, Apache 2.0).
- L3Cube HingCorpus, HingBERT and HingLID; GLUECoS benchmark (code-mixed NLP).
- SisFall, MobiAct, UP-Fall (accelerometer fall datasets; none contain real falls).
- PhysioNet open datasets (vital-sign dynamics; some need credentialed access).

Open questions. Real call-note formats at Samarth? Escalation rules for serious alerts? Preferred telephony provider? Data-residency requirement?